Luis Enrique Dalmau Coopat

AI Engineer · Computer Vision · Full-Stack

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Professional Summary

AI/ML Engineer with 3+ years designing, optimizing, and deploying production computer-vision systems, from dataset tooling to edge inference on industrial hardware. Author of a proprietary computer-vision library used to train and deploy YOLOv9 object-detection models for IIoT predictive maintenance, covering 10+ detection architectures. Experienced in MLOps, model optimization for deployment (TensorRT, TFLite), and full-stack delivery with FastAPI, React, and Docker. B.S. in Computer Science (University of Havana) and former instructor in AI, ML, and distributed systems. Spanish native, English B2 (professional working proficiency).

Skills

AI & Computer Vision: Python · PyTorch · TensorFlow · OpenCV · Deep Learning · Object Detection · Instance Segmentation · Multi-Object Tracking · Transfer Learning · Model Evaluation (mAP, precision/recall) · Auto-labeling (GroundingDINO, SAM, Florence-2) · Synthetic Data (Stable Diffusion) · Albumentations · TensorBoard · scikit-learn · pandas · NumPy

Generative AI: LLM integration (Google Gemini) · RAG pipelines (LangChain) · Prompt engineering

Backend: Python · C++ · FastAPI · SQLAlchemy 2.0 · Alembic · Django · Pydantic · REST APIs · JWT/OAuth2 · APScheduler

Frontend: React 18 · Next.js 14 · TypeScript · Tailwind CSS · TanStack Query · Formik · Chart.js

Databases: PostgreSQL · MongoDB · MySQL · Redis

MLOps & DevOps: Docker · Docker Compose · ONNX export · TensorRT · TFLite · Git · GitLab CI · GitHub Actions · AWS S3 · Heroku · Nginx · Linux

Testing & Quality: pytest · pytest-asyncio · pytest-cov · moto · httpx · ESLint · Prettier

Experience

Technical Engineer

Anukys · Málaga, Spain (Hybrid) | 10/2025 – Present

- Deliver AI and computer-vision projects for client companies and business partners, from model deployment to technical advisory.

AI Engineer — Computer Vision

SORBA.ai (Sorbotics LLC) · Jacksonville, USA (Remote) | 01/2024 – Present

- Built and optimized CV training pipelines covering 10+ architectures (YOLOv5/v9, EfficientDet, MobileNet v2, RetinaNet, Faster R-CNN, Mask R-CNN, CenterNet, SSD).
- Shipped these capabilities across 22 company projects, from dataset preparation to on-plant deployment.
- Integrated synthetic data generation (Stable Diffusion, GAN, VAE), auto-labeling (GroundingDINO, Florence-2), and model optimization (TensorRT, TFLite).
- Built real-time video processing for industrial detection: fire/smoke, gas leaks, PPE, gauge OCR, license plates, and spatial tracking.
- Packaged ML models for Debian/RHEL. Managed GPU resources for concurrent training.

Full-Stack Developer (Backend focus)

Aerial Underground Construction Inc. · Kissimmee, FL, USA (Remote, concurrent with SORBA.ai) | 09/2024 – 10/2025

- Built logistics platform backend (FastAPI + SQLAlchemy + Alembic + PostgreSQL): JWT auth, RBAC (3 roles), 20+ REST endpoints.
- Implemented commissions, daily/weekly/monthly reports, PDF/Excel invoices (ReportLab/openpyxl), AWS S3 (boto3), scheduled jobs (APScheduler).
- Developed ~35% of the frontend codebase (Next.js 14 + React + TypeScript) with automated tests and Docker/Heroku deployment.

Assistant Professor

University of Havana · Faculty of Mathematics & Computer Science (part-time, concurrent) | 01/2023 – 01/2025

- Taught Computer Networks, Machine Learning, AI, Operating Systems, and Distributed Systems; led practicals, graded exams, mentored students.

Projects

Computer Vision Library (proprietary) — author: YOLOv9 training, auto-labeling, synthetic data, GPU management.

Logistics Management System — FastAPI + Next.js + PostgreSQL; RBAC, reports, invoices, AWS S3.

YOLO-3D — Real-time 3D object detection: YOLOv11 + Depth Anything v2.

Image Analyzer (BLIP) — Multimodal image analysis: descriptions, keywords.

LicitAI — FastAPI + React + MongoDB + Google Gemini (LLM) + Stripe.

Protectora RRSS — Pet-adoption diffusion: FastAPI + React + Telegram bot + SMTP.

ML Trainer Engine — CV training with SOTA models (RF-DETR, EfficientDet, MobileNet, DeepLabV3).

Education & Certifications

B.S. in Computer Science — University of Havana (2017–2022). Thesis: "Graphic generation of signs of the Cuban Sign Language".

CS50's Introduction to AI with Python — HarvardX (edX) · **Data Science & Machine Learning** — Sololearn.

Languages

Spanish: Native · **English:** B2 (Professional Working Proficiency)